Titanic Dataset – Exploratory Data Analysis

Data Analyst Internship | Task 5

1. Objective

The objective of this project is to explore the Titanic passenger dataset using Python, Pandas, Matplotlib and Seaborn, and identify patterns, trends, relationships and missing data through statistical summaries and visualizations.

2. Dataset Overview

The dataset contains 891 passenger records and 12 columns. It includes passenger details such as survival status, passenger class, name, gender, age, family relationships, ticket fare, cabin and port of embarkation.

Dataset measure	Result
Rows	891
Columns	12
Duplicate rows	0
Male passengers	577
Female passengers	314
Survived	342
Did not survive	549
Third class passengers	491
Second class passengers	184
First class passengers	216

3. Data Quality and Missing Values

No duplicate rows were found. Missing values were present in three columns: Age (177), Cabin (687) and Embarked (2). Cabin contains the largest number of missing values. The notebook reports these missing values; it does not apply an imputation or deletion strategy.

4. Visualizations Performed

Visualization	Purpose
Survival count	Compare the number of passengers who survived and did not survive.
Gender distribution	Compare the number of male and female passengers.
Passenger class distribution	Show passenger counts in classes 1, 2 and 3.
Age histogram with KDE	Explore the distribution of passenger ages.
Fare histogram with KDE	Explore ticket fare distribution and its skew.
Age boxplot	Identify the spread of ages and potential outliers.
Survival by gender	Compare survival counts across genders.
Survival by passenger class	Compare survival counts across passenger classes.
Age vs fare scatterplot	Explore age and fare together, with survival as hue.
Correlation heatmap	Review pairwise correlations among numerical columns.
Pairplot	Explore pairwise patterns among selected numerical features, coloured by survival.

5. Key Findings

• The dataset has 891 rows, 12 columns and no duplicate records.
• There are more male passengers (577) than female passengers (314).
• 342 passengers survived and 549 did not survive; the observed survival proportion is approximately 38.4%.
• Third class is the largest passenger group, with 491 passengers.
• The survival-by-gender chart shows a higher survival count for females than males; counts should not be confused with within-group survival rates.
• The survival-by-class chart indicates that first-class passengers generally had better survival outcomes than third-class passengers.
• The fare histogram is right-skewed, with a smaller number of high-fare observations.
• Age, Cabin and Embarked contain missing values, with Cabin having the most.

6. Conclusion

Exploratory data analysis revealed useful patterns in passenger composition and survival outcomes. The visual analysis highlights differences by gender and passenger class, while the statistical inspection identifies missing data that should be considered before further modelling. The notebook uses descriptive statistics, count plots, histograms, a boxplot, a scatterplot, a correlation heatmap and a pairplot to explore the dataset.

7. Tools Used

Python • Pandas • NumPy • Matplotlib • Seaborn • Google Colab

Prepared as the PDF findings report for Data Analyst Internship – Task 5.